Energy yield loss during communication outage

Wind Farm T (demo) · 01.09.2026 – 03.09.2026 · turbines T1, T2, T3

ESTIMATED LOST ENERGY (P50)

313.6 MWh

LIKELY RANGE (P10 – P90)

273.1 – 368.1 MWh

Over 72 h for 3 turbines. Value at 73.50 EUR/MWh: EUR 23,046 (EUR 20,075 – 27,057).

1 Outage details

Wind farm	Wind Farm T (demo)
Operator / insured	Demo Operator GmbH
Insurer · policy	Demo Insurance AG · DEMO-0000-00
Outage start	01 September 2026 at 00:00 (local time, Europe/Berlin)
Outage end	04 September 2026 at 00:00 (local time, Europe/Berlin)
Duration	72 hours
Affected turbines	T1, T2, T3 (4.5 MW class)
Suspected cause	A production stop is suspected; possible causes include cable damage between the wind farm and the grid connection point or a technical fault preventing feed-in. The root cause is under investigation.
Waiting period	24 h → 191.1 MWh (P50) beyond the waiting period, EUR 14,043

2 Lost energy per turbine

Turbine	P10 MWh	P50 MWh	P90 MWh	Avg wind m/s	Avg power MW
T1	72.2	89.2	104.4	5.8	1.24
T2	116.5	126.2	149.6	6.7	1.75
T3	84.4	98.2	114.1	6.2	1.36
Wind farm	273.1	313.6	368.1		

P50 is the best estimate; actual lost energy is expected to fall between P10 and P90 in 8 out of 10 cases, based on the method's historical errors. Farm P10/P90 are summed per turbine (conservative, since all turbines share the same reference data). Power curve values are interpreted as kW; energy = expected power × 10 min.

3 Reconstructed wind and lost production



Figure 1 – Measured wind speed at each turbine (navy) one day before and after the outage, and the reconstructed wind speed during the outage (shaded) with its P10–P90 band.

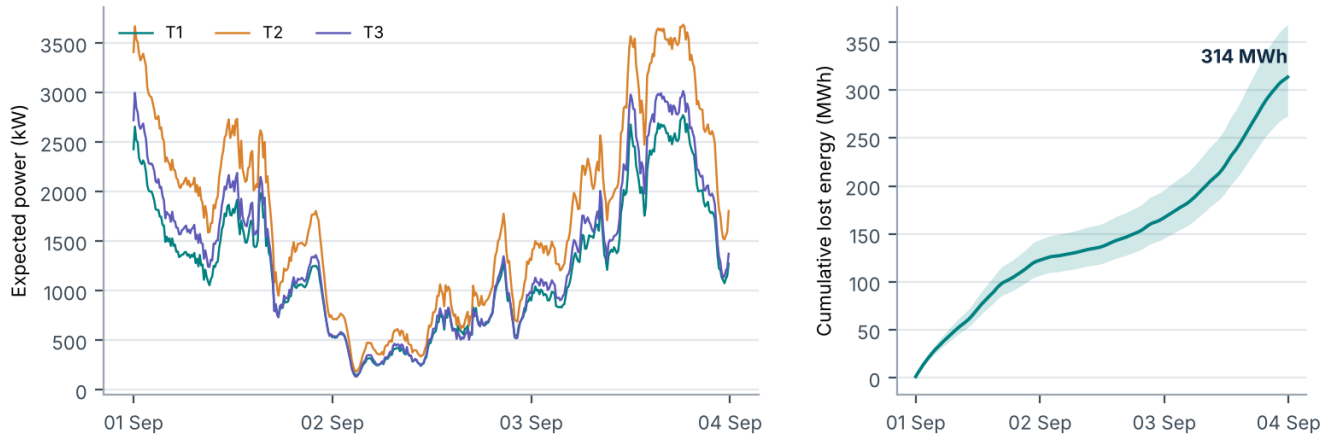


Figure 2 – Left: expected power per turbine during the outage. Right: cumulative lost energy for the wind farm with P10–P90 band.

4 Method, accuracy and limitations

Wind speeds at the affected turbines were reconstructed from 7 reference turbines in park(s) H, O outside the affected wind farm. A linear regression, fitted on all available history outside the outage, relates each target turbine's wind speed to the park-average reference wind speed (including values up to 30 minutes before and after each interval) and the reference wind direction. Lost energy is the turbine power curve applied to the reconstructed wind, averaged over the model's historical error distribution.

Why this method: the relationship is linear and can be checked by hand – each target turbine's wind is a weighted combination of the reference parks' wind, adjusted for wind direction. In tests it was as accurate as a more complex machine-learning model, so the simpler, auditable method was chosen.

Turbine	Wind RMSE	Wind MAE	Power RMSE	|Energy err.| %	Val. wind RMSE	Val. energy %
T1	1.23	0.93	679	12.1	1.54	+13.4
T2	1.10	0.84	630	8.0	1.17	-0.6
T3	1.10	0.84	606	8.8	1.28	-1.6

Backtest columns: average over 27 simulated 3-day outages across the data period (model refitted without each window). 'Val.' columns: held-out validation week. Wind in m/s, power in kW, energy error = deviation of summed energy.

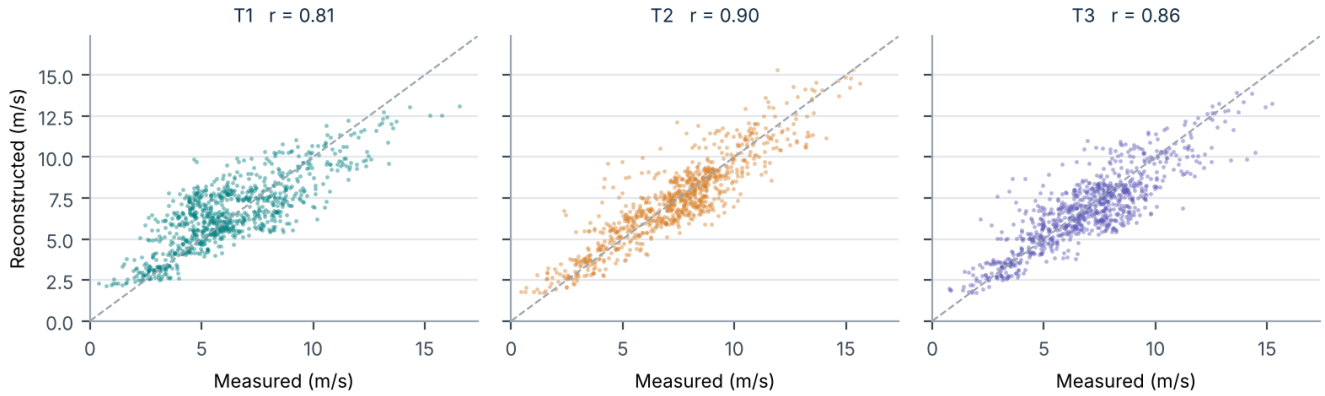


Figure 3 – Validation week: reconstructed vs measured 10-minute wind speed. Points on the dashed line would be perfect predictions.

Limitations

• The figure is an estimate. Individual 10-minute values are uncertain; totals over several days are considerably more reliable.
• It assumes the turbines were available and would have operated normally; curtailment, maintenance or grid limits during the outage are not included.
• Loss of communication does not by itself prove a production stop. The grid operator's metering data for the period should be used to confirm that no energy was fed in.
• Monetary value, waiting period and policy terms depend on contracts provided by the operator.

Appendix A Reproducibility and audit trail

Re-running the tool with the input files below (verified by their SHA-256 fingerprints) and the same model version reproduces every figure in this report.

Model	Park-mean linear MCP with lags and direction (v0.1.0-baseline)
Reference turbines	H1, H2, H3, H4, H5, O1, O2
Outage window	2026-09-01 00:00 – 2026-09-03 23:50 (Europe/Berlin, 10-min stamps)
Power curve	rated 4,500 kW from 11.5 m/s; first output at 3 m/s
Report generated	24 Oct 2026 11:32

Turbine	Weight park H	Weight park O	Intercept (m/s)	R^2	Training rows
T1	0.46	0.41	0.35	0.65	12372
T2	0.46	0.41	0.29	0.76	12419
T3	0.46	0.34	0.45	0.71	12417

Weight = total regression coefficient on a reference park's mean wind speed (summed over the time-shifted inputs). Example: weight 0.45 means +1 m/s at that park adds about 0.45 m/s at the target turbine.

Input file	SHA-256 fingerprint
H1_data.xlsx	`f00af7198126ac3a5839ab4cb1a09eb6ec54bf2bc98f7485b7ccd1afbf1334c5`
H2_data.xlsx	`5dfa885703128edcdddf2b4c7729f22219fd5803648f27c6a73737024bea0365`
H3_data.xlsx	`ddb005bfaf46af048cd5e752c63c78fae566c20f786be152872ba5178b3e2c4b`
H4_data.xlsx	`f4a213f36e8400ec36d444d2e53ef01d67658384f2078a4c8c334c861364caf6`
H5_data.xlsx	`ffa19bb61b99c3ae5347dac1c671e20955c70f466d1851466edf7b58cfc1722e`
O1_data.xlsx	`7139e3380bb1f1acc4f9094a7e372f03f50d7fbd06178c2a4ac269a7e541bfd7`
O2_data.xlsx	`d12d19e7a0c496c213cb24ce0a88c9290b212404b70b21b437a021857d0ea292`
T1_data.xlsx	`56e4139af0334f3e28393ac5757b545323790fc89043b0d4cbda08d400731b55`
T1_power_curve.xlsx	`cd7fbbf142148d9fcd84e95cfb9e2404c2e9a9bd399d13cf3dc669a4e123a6c1`
T2_data.xlsx	`3a6bb6e248a8ce67fd224a85ac433b09b48d89479b6044a92f43569f9a76b9e3`
T2_power_curve.xlsx	`2cfcffb092bb7412deb12ee0332b4e7d8200ea9768f211cd0961c18ee03f389e`
T3_data.xlsx	`2aba8062d668474ace743c0fc4c79c802137af72d63f43edd0d91b3e30d5fc4a`
T3_power_curve.xlsx	`7434f64acf9b77bfefd084f4b26d25a0cd86a136206bf21048ef4548a50a6f02`